THE LAWS OF SOFTWARE TEAMS

Murphy's law

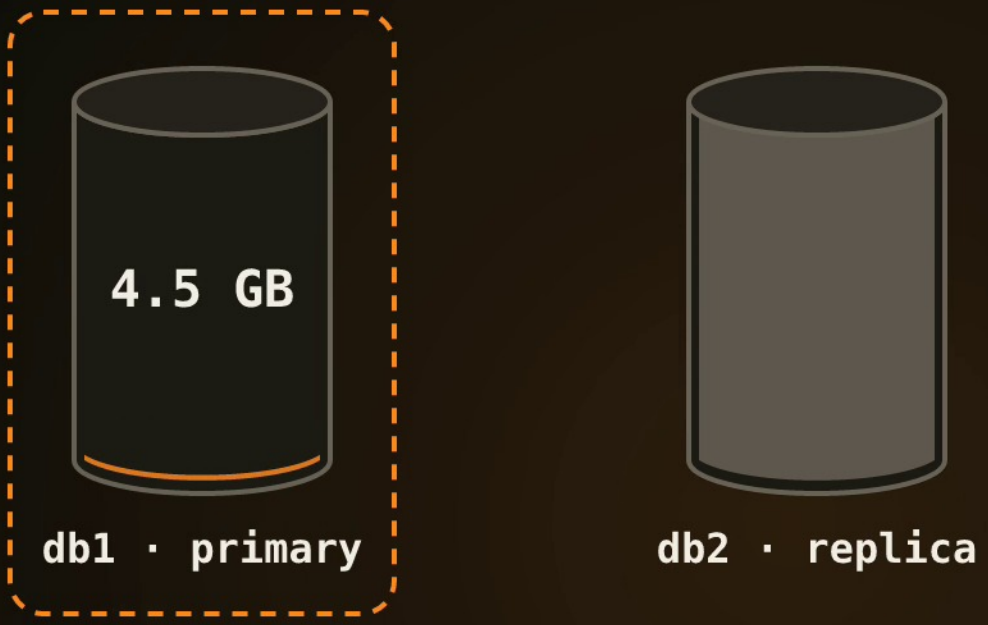
Anything that can go wrong will go wrong

swipe →

Ivan Rykov

GitLab, January 2017

Wrong server




```
$ rm -rf …/postgresql/data
```

~300 GB of production data, gone

5 backup methods
0 working reliably

LVM snapshots
only once every 24 hours

Regular backups
files a few bytes in size

pg_dump
failed silently

Azure disk snapshots
not enabled for the DB servers

S3 backups
bucket was empty

6 hours of data lost for good

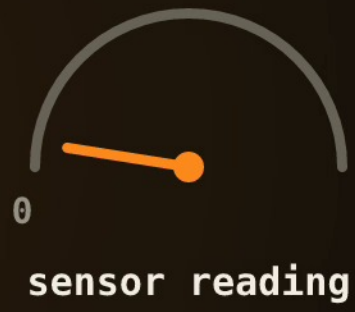
GitLab.com database incident postmortem, Feb 1, 2017

03 THE LAW

Anything that can go wrong will go wrong

Edward A. Murphy Jr. · US Air Force · c. 1949

rocket-sled test



Sensors wired backwards

the exact origin story is disputed

04 THE MATH

Not pessimism
Math

A bug with a 1 in 1,000 chance per deploy

```
deploys: 2600
```

93%

chance it happens at least once

```
1 - 0.999^2600 ≈ 0.93
```

It happens to the best

ARIANE 5 · 1996

64-bit number, 16-bit slot



16 bits

Overflow → destroyed after 37 seconds

KNIGHT CAPITAL · 2012



over **$460M** lost in 45 minutes

Netflix, 2011:

Chaos Monkey

switches off its own servers in production



On purpose

So failure stops being a surprise

Plan for it

1. **Test restores**
   not just backups

2. **Roll out in stages**
   one server first, then the rest

3. **“That’ll never happen”**
   → check it anyway

via manager.dev

08 TAKEAWAY

Murphy isn't a curse
It's a checklist

- Restores tested ✓
- Rollouts staged ✓
- Failures rehearsed ✓
- "Never happens" checked ✓

the last law in this series

TAKEAWAY

Murphy isn't a curse
It's a checklist

Sources: GitLab, 2017 · Ariane flight V88, 1996 · SEC order on Knight Capital, 2013 · Netflix, 2011

next: the series recap